Worksheet & Answer Key

Carbon compounds can undergo substitution reactions, which involve replacing one functional group with another. In this type of reaction, a _____ (1) molecule attaches to a carbon atom in place of an existing group. This process often requires the presence of a _____ (2) and may result in the formation of a new compound with unique properties. Substitution reactions are important in fields like _____ (3), where they are used to synthesize new materials. Chemists must carefully control reaction conditions, such as temperature and catalysts, to achieve successful _____ (4). The resulting compounds can have improved properties, making them useful for a variety of _____ (5) applications.

Word Bank:

molecule, presence, fields, conditions, catalysts

Fill in the blanks:

1. Carbon compounds can undergo substitution reactions, which involve replacing one functional group with another. In this type of reaction, a _____ molecule attaches to a carbon atom in place of an existing group.

2. This process often requires the presence of a _____ and may result in the formation of a new compound with unique properties.

3. Substitution reactions are important in _____ where they are used to synthesize new materials.

4. CHEMISTS must carefully control reaction _____, such as temperature and catalysts, to achieve successful substitution.

5. The resulting compounds can have improved properties, making them useful for a variety of _____ applications.

■ Answer Key:

1. molecule

2. presence

3. fields

4. conditions

5. catalysts